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REVIEW ARTICLE

Detecting dental caries on oral photographs using artificial intelligence: A systematic review

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Abstract

Objectives: This systematic review aimed at evaluating the performance of artificial intelligence (AI) models in detecting dental caries on oral photographs.

Methods: Methodological characteristics and performance metrics of clinical studies reporting on deep learning and other machine learning algorithms were assessed. The risk of bias was evaluated using the quality assessment of diagnostic accuracy studies 2 (QUADAS-2) tool. A systematic search was conducted in EMBASE, Medline, and Scopus.

Results: Out of 3410 identified records, 19 studies were included with six and seven studies having low risk of biases and applicability concerns for all the domains, respectively. Metrics varied widely and were assessed on multiple levels. F1-scores for classification and detection tasks were 68.3%–94.3% and 42.8%–95.4%, respectively. Irrespective of the task, F1-scores were 68.3%–95.4% for professional cameras, 78.8%–87.6%, for intraoral cameras, and 42.8%–80% for smartphone cameras. Limited studies allowed assessing AI performance for lesions of different severity.

Conclusion: Automatic detection of dental caries using AI may provide objective verification of clinicians' diagnoses and facilitate patient-clinician communication and teledentistry. Future studies should consider more robust study designs, employ comparable and standardized metrics, and focus on the severity of caries lesions.

KEYWORDS
deep learning, dental caries, intraoral camera, oral photograph, smartphone

1 | INTRODUCTION

Approximately two billion adults and 520 million children suffer from dental caries in permanent and primary teeth, respectively. Dental caries is largely preventable, but if untreated, constitutes a major health burden. The prevalence of dental caries continues to increase in most low- and middle-income countries, which disproportionately affects marginalized and underserved groups (World Health Organization, 2022) Furthermore, dental facilities and providers are often unevenly geographically distributed, increasing barriers to access to dental care (Peres et al., 2019). Early caries detection coupled with improved hygiene behaviors and preventive interventions can arrest lesions and reduce treatment time and costs (Twetman, 2018).

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In daily clinical practice, caries detection is first and foremost based on visual-tactile examination. However, clinical examination is subjective since the accuracy of caries diagnosis depends on the experience of clinicians as well as a range of clinical factors (Diniz et al., 2012). To reduce the variability of visual-tactile examinations, auxiliary diagnostic methods such as digital photography have received attention (Boye et al., 2012, 2013). Intraoral photographs are considered a straightforward and non-ionizing image acquisition technique, and several studies demonstrated their feasibility and promises for supporting caries diagnosis (Boye et al., 2012; Umemori et al., 2010).

Artificial intelligence (AI) models have been suggested to assist with the diagnosis of dental conditions. Recent implementations of AI for imaging rely on deep learning (DL), a subfield of machine learning (ML) (Morid et al., 2021). DL departed from earlier ML approaches primarily by replacing human-engineered features with high-capacity neural networks trained using large amounts of data allowing automated feature extraction. The first applications of DL in dentistry evolved around radiographic image analysis, for example, for detecting dental caries (Lee et al., 2018a) or periodontal bone loss (Lee et al., 2018b).

When evaluating AI-based studies in dentistry, a distinction should be made between image classification, detection, and segmentation. Image classification refers to assigning a categorical label, usually binary, to the entire image (e.g., sound/caries). Detection refers to determining the area of the outcome measure with bounding boxes and can be considered as the combination of classification and localization. Segmentation refers to providing the exact outline for the outcome measure and separating it from the rest of the image using pixel-wise annotations (Khan & Al–Habsi, 2020).

The field of dentistry has seen a steady rise in publications concerning the application of AI models. A recent systematic review analyzed the use of DL in diagnosing dental caries through various image sources such as radiographs and near-infrared light transillumination (Mohammad-Rahimi et al., 2022). In contrast, our review focuses on comparing different AI models (both DL and non-DL) for detecting caries lesions solely through oral photographs. We also investigate the impact of various image acquisition devices such as professional, intraoral, and smartphone cameras as well as lesion severity. Our review aims to address the limitations of this new line of research and shed light on the chances and challenges of using AI in detecting caries from oral photographs for clinical practice, teledentistry, and public screening.

2 | MATERIALS AND METHODS

2.1 | Protocols

This systematic review was prepared according to the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) statement (Moher et al., 2009). The focused question was addressed using the participant, intervention, comparison, and outcomes (PICO) criteria (Miller & Forrest, 2001; Richardson et al., 1995). The study protocol was registered on the international prospective register of systematic reviews (PROSPERO) platform under the registration number CRD42023410147.

2.2 | Search strategy

Three electronic databases (Medline, Scopus, and EMBASE) were used to identify publications that met the inclusion criteria. The search was conducted up to December 15, 2022, using the following combination of MeSH terms and keywords: (detect* OR segment* OR classif* OR recognize OR recognition OR diagnosis OR differentiat* OR discriminat* OR risk OR explanation OR mark*) AND (caries OR oral lesion OR tooth lesion OR teeth lesion OR decay OR white spot OR cavity) AND (artificial intelligence OR AI OR machine learning OR deep learning OR neural network* OR supervised learning OR semisupervised learning OR unsupervised learning OR computer vision OR multilayer perceptron OR MLP). Besides the electronic databases, reference lists of the selected studies were manually screened.

2.3 | Eligibility criteria

The following inclusion criteria were used to screen and assess the retrieved records:

1. Being an original peer-reviewed article or conference proceedings in English.
2. In vivo study.
3. Reporting on dental caries of the human tooth.
4. Reporting on the diagnosis of dental caries using DL and non-DL models.
5. Using oral photographs as the data source.
6. Reporting on the performance of the DL and non-DL models.

Records were excluded if they were:

1. Animal studies.
2. Based on data sources other than oral photographs.

2.4 | Focused PICO question

Can AI-based models be used for detecting dental caries lesions on oral photographs?

Participants: digital intraoral photographs from the oral cavity and teeth.

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Intervention: application of AI-based models (DL and non-DL) for caries detection.

Comparison: reference standards such as clinical dental examinations or image annotations by dental professionals.

Outcomes: performance metrics such as sensitivity, specificity, precision, F1-score, accuracy, area under receiver operating characteristics (AUROC).

2.5 | Selection of studies

A two-stage screening, title and abstract followed by full-text reading, was carried out by two authors independently (MM, JF). Title management was performed by a commercially available software program (Covidence systematic review software, Veritas Health Innovation). The duplicates were removed within and between the databases. The full texts of potential articles were retrieved and evaluated using the eligibility criteria. Any disagreements on the selection of studies were discussed and resolved. The reasons for excluding articles not meeting the eligibility criteria were reported.

2.6 | Data extraction

Using a predesigned data extraction form, the following information was extracted from the papers that met the eligibility criteria: title, authors' names, contact address, study location, year of publication, study design, sample size, frequency of caries, ground truth for diagnosis, tooth surface, type of dentition (permanent vs primary), region of interest (anterior vs. posterior), presence of restorations, presence of plaque/calculus, image acquisition device, type of DL and non-DL models, sample allocation (training, validation, test), task formulation (classification, detection, segmentation), performance measures (e.g., sensitivity, specificity, precision, F1-score, accuracy, AUROC), and authors' conclusion.

2.7 | Risk of bias

The Quality assessment was conducted using the quality assessment of diagnostic accuracy studies (QUADAS-2) criteria independently and in duplicate by two authors (MM and JF) (Whiting et al., 2011). The QUADAS-2 checklist consists of four risks of bias domains, namely, data selection, index test, reference standard, and flow and timing, and three domains evaluating applicability concerns of a study on patient selection, index test, and reference standard. In this current review, the modified version of QUADAS- 2 by Mohammad-Rahimi et al. (2022) was used. The final implemented criteria are tabulated in Table A1. If all signaling questions for a domain were answered "yes," then the risk of bias was judged as "low"; if any of the signaling questions were answered no, then the risk was recorded as "high." The risk of bias was recorded as "unclear" only when insufficient data were presented.

2.8 | Data analysis

Meta-analysis was not possible due to considerable methodological and clinical heterogeneity between the included studies. We recorded lesion prevalence, caries classification system, tooth surface and type, image acquisition device, presence of plaque/calculus, presence of restorations, and analysis task (i.e., classification, detection, segmentation).

Our narrative synthesis focused on a range of metrics, namely sensitivity/recall, specificity, precession, F1, accuracy, AUROC, and negative predictive value (NPV). The following formulae were used for calculating metrics according to true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN):

$$\text{Accuracy: } (TP + TN) / (\text{Total number of samples})$$

$$\text{Specificity: } TN / (TN + FP)$$

$$\text{Sensitivity / Recall: } TP / (TP + FN)$$

$$\text{Precision: } TP / (TP + FP)$$

$$F1 - \text{Score: } 2 \times (\text{Precision} \times \text{Sensitivity}) / (\text{Precision} + \text{Sensitivity})$$

For detection tasks, we also reported on mean average precision (mAP) as well as the Intersection over Union (IoU) threshold based on which classification metrics were calculated. In our synthesis, where available or could be manually calculated, we reported mainly the F1-score, which provides a harmonized average of recall/sensitivity and precision. If it was not available, we reported on AUROC or both sensitivity and specificity. Note that some studies reported accuracy, which is prone to mainly reflect the prevalence of caries lesions and not necessarily model performance (in the case of heavily imbalanced datasets, which are common); we do not report on results solely based on accuracy in our synthesis.

Performance metrics were stratified based on the type of AI task, image acquisition device, and severity of caries lesions across the included studies. If the performance metrics were not available based on the test sets, metrics on the validation set were recorded in the tables, but only the results from the test sets were used to evaluate and compare models.

3 | RESULTS

3.1 | Study selection and search results

The systematic search resulted in 3410 unique records. After screening the title and abstracts 3380 records were excluded. From the remaining 30 records, 13 articles were excluded in the full-text reading stage: four articles were in vitro studies (Berdouses et al., 2015; Duong, Kabir & Kuo, 2021, Duong, Nguyen et al., 2021; Ghaedi et al., 2014), four articles had

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different outcome measures than dental caries (Eto et al., 2022; Li et al., 2021; Muramatsu et al., 2022; Takahashi et al., 2021), three articles used different imagery data from oral photographs (Casalegno et al., 2019; Holtkamp et al., 2021; Schwendicke et al., 2020), one article used artificial caries (Kositbowornchai et al., 2006), and one article did not report on the performance measures (Guijarro-Rodríguez et al., 2020). After manually screening the reference lists, two other studies were added, and the 19 included studies were ultimately processed for data extraction (Askar et al., 2021; Ding et al., 2021; Javid et al., 2020; Jiang et al., 2021; Khan et al., 2022; Kim et al., 2021; Kühnisch et al., 2022; Liu et al., 2020; Moutselos et al., 2019; Navarro et al., 2019; Park et al., 2022; Rashid et al., 2022; Saini et al., 2021; Sonavane et al., 2021; Thanh et al., 2022; Wang et al., 2020; Yu et al., 2020; Zhang et al., 2020, 2022). The summary of the search strategy is depicted in the PRISMA flowchart in Figure 1.

3.2 | Assessment of methodological quality

Detailed information on the risk of bias and applicability concerns are shown in Table 1 and Figure 2. Regarding the risk of bias, six studies were considered to have a low risk for all four domains of QUADAS-2 (Askar et al., 2021; Kim et al., 2021; Kühnisch et al., 2022; Park et al., 2022; Yu et al., 2020; Zhang et al., 2022). Regarding applicability concerns, seven studies were categorized as having low concern for all three domains (Askar et al., 2021; Jiang et al., 2021; Kim et al., 2021; Park et al., 2022; Thanh et al., 2022; Zhang et al., 2020, 2022).

Regarding the risk of bias for patient selection, seven, two, and 10 studies had low, high, and unclear risks of bias, respectively. For the reference standard, 14 and five studies had low and high risks of biases, respectively. For index test, 17 and two studies had low and high unclear risks of bias, respectively. For flow and timing, 17 and two studies had a low and high risk of bias, respectively.

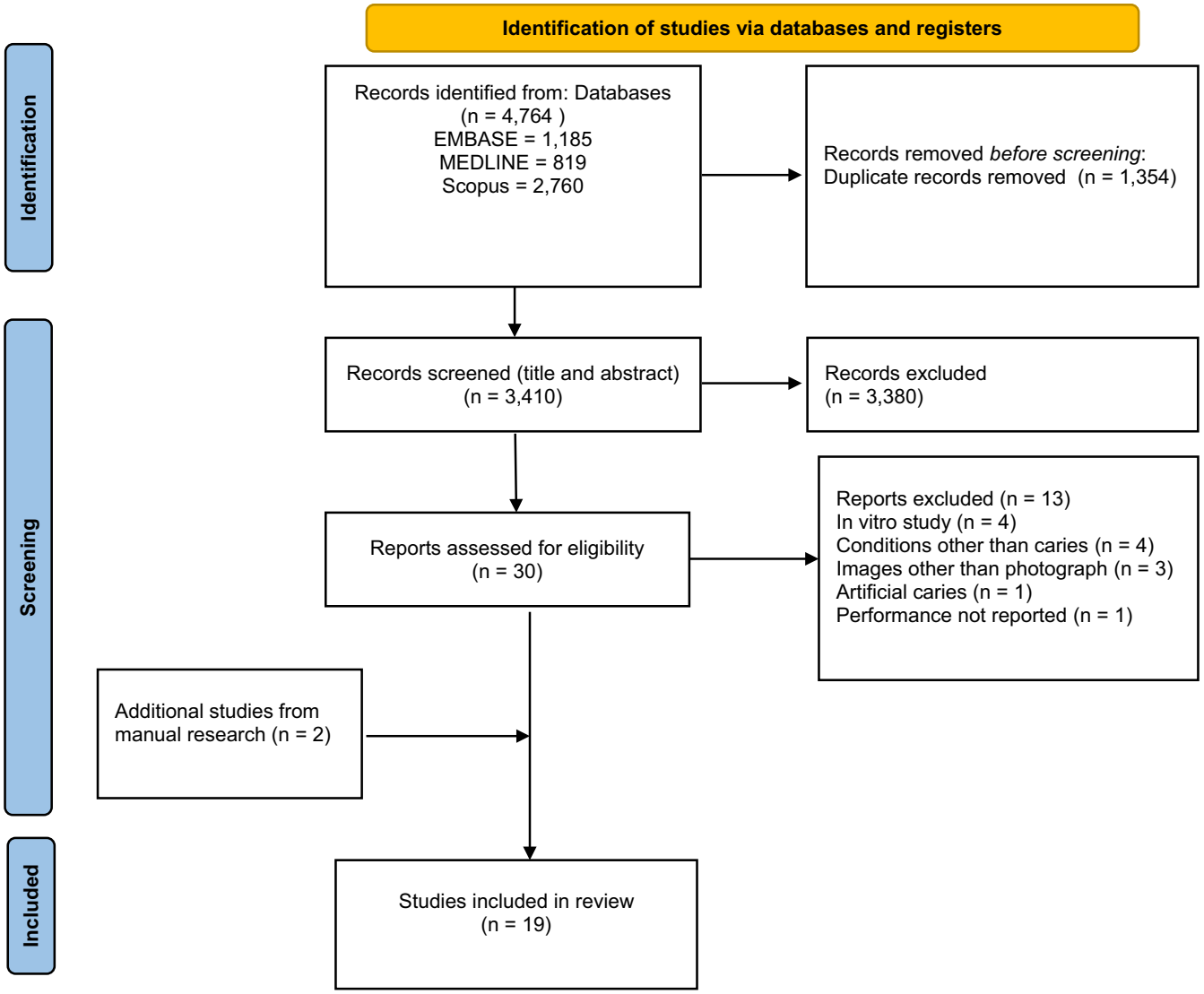


FIGURE 1 The flowchart of the search process. From: Page et al. (2021).

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TABLE 1 Risk of bias and applicability concerns of included studies.

Study	Risk of bias				Applicability concerns		
	Patient selection	Index test	Reference standard	Flow and timing	Patient selection	Index test	Reference standard
Moutselos et al. (2019)	Unknown	Low	Low	Low	High	Low	Low
Navarro et al. (2019)	Unknown	Low	High	Low	Unknown	High	High
Zhang et al. (2020)	Unknown	Low	Low	Low	Low	Low	Low
Wang et al. (2020)	Unknown	Low	Low	Low	High	Low	Low
Liu et al. (2020)	Low	High	High	High	Low	High	High
Javid et al. (2020)	Unknown	Low	Low	Low	Low	Low	High
Yu et al. (2020)	Low	Low	Low	Low	High	Low	Low
Sonavane (2021)	High	Low	High	Low	High	High	High
Kim et al. (2021)	Low	Low	Low	Low	Low	Low	Low
Askar et al. (2021)	Low	Low	Low	Low	Low	Low	Low
Ding et al. (2021)	Unknown	Low	Low	Low	Low	Low	Unknown
Jiang et al. (2021)	Unknown	Low	Low	Low	Low	Low	Low
Saini et al. (2021)	Unknown	High	High	Low	High	High	High
Zhang et al. (2022)	Low	Low	Low	Low	Low	Low	Low
Kühnisch et al. (2022)	Low	Low	Low	Low	High	Low	Low
Rashid et al. (2022)	Unknown	Low	Low	Low	Unknown	Low	High
Park et al. (2022)	Low	Low	Low	Low	Low	Low	Low
Thanh et al. (2022)	Unknown	Low	Low	Low	Low	Low	Low
Khan et al. (2022)	High	Low	High	High	High	High	High

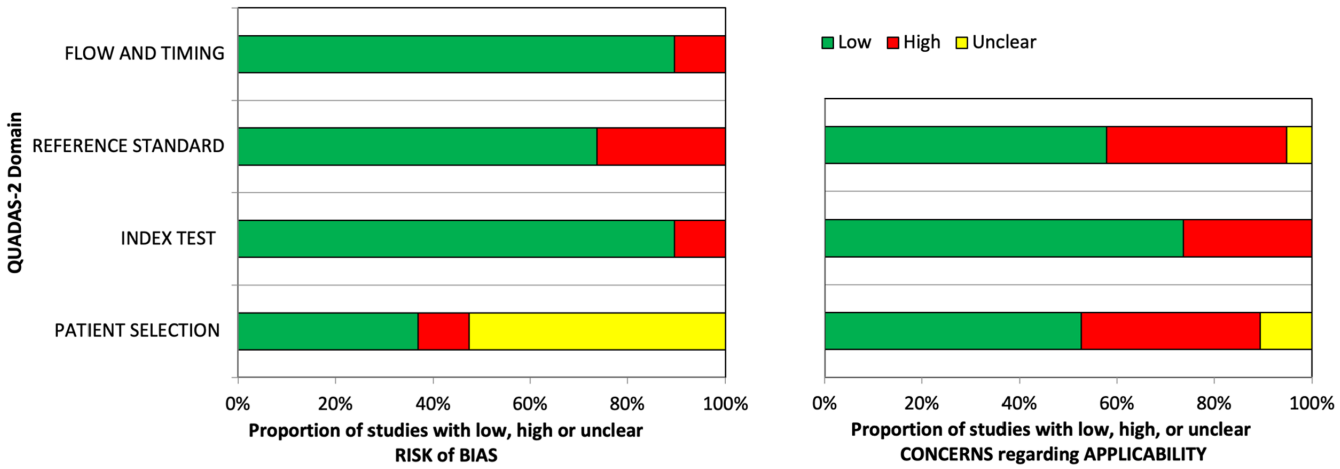


FIGURE 2 The proportion of studies with low, high, and unclear risk of bias or applicability concerns.

Regarding applicability concerns for patient selection, 10, seven, and two studies raised low, high, and unclear concerns, respectively. For index test applicability, 14, and five raised low and, respectively. For reference standard applicability, 11, seven, and one studies raised low, high, and unclear concerns, respectively.

3.3 | Study characteristics

The characteristics of the included studies are summarized in chronological order in Table 2. The publication dates ranged from 2019 to 2022. Overall, for the 19 included studies, sample sizes ranged from 45 to 12,600 with 10 studies having sample sizes larger than 1000. The prevalence of caries lesions was reported in nine studies and varied significantly, with some studies having more sound than carious teeth (Askar et al., 2021; Kim et al., 2021; Liu et al., 2020; Park et al., 2022; Yu et al., 2020; Zhang et al., 2022) and vice versa in others (Jiang et al., 2021; Kühnisch et al., 2022; Sonavane et al., 2021). Overall, only six studies used ICDAS, either the full or merged system, as a reference standard (Kühnisch et al., 2022; Moutselos et al., 2019; Park et al., 2022; Thanh et al., 2022; Zhang et al., 2020, 2022).

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TABLE 2 Characteristics of the included studies in chronological order.

References	Population	Sample size	Train vs. test (ratio/number)	Frequency of caries	Reference standard (ICDAS vs. non-standard)	Ground truth for caries diagnosis	Tooth surface	Restoration (yes/no)	Plaque/ calculus (yes/ no)	Type of dentition (primary/ permanent)	Tooth location
Moutselos et al. (2019)	Greece	Images: 88	*Training* 79 (N) *Test* 9 (N)	NM	Full ICDAS: 0 (sound) to 6 (extensive distinctive cavity with visible dentine)	Visual Annotation	Occlusal	NM	No	Permanent	Posterior
Navarro et al. (2019)	Philippines	Images: 45	*Training* 30 (N) *Test* 15 (N)	NM	Non-standard: Sound Caries	Visual Annotation	Smooth	NM	NM	NM	Posterior/ Anterior
Zhang et al. (2020)	USA	Images: 1000	*Training* 80% *Test* 20%	NM	Merged ICDAS: Sound Non-cavitated Cavitated Others (treatments)	Visual Annotation	Smooth	NM	NM	Primary and permanent	Posterior/ Anterior
Wang et al. (2020)	China	Images: 7200	*Training* 80% *Test* 20%	NM	Non-standard: Sound White-spot lesions Smashed	Visual Annotation: More than two dentists	Occlusal, Smooth	NM	Yes	NM	Posterior/ Anterior
Liu et al. (2020)	China	Images: 12,600	*Training* 12,600 images (N) *Test* 100 subjects (N)	Caries (yes): 2556	Non-standard: Sound Caries	NM	NM	NM	Yes	NM	Posterior/ Anterior
Yu et al. (2020)	China	Images: 1638 Participants: 342	*Training* Images: 1052 (N) Participants:242 (N) *Test* Images: 316 (N) Participants: 100 (N)	Sound: 1451 Caries: 187	Non-standard: Sound Caries	Visual Annotation: Two expert dentists	NM	NM	Yes	First permanent molar in children	Posterior
Javid et al. (2020)	Pakistan	Images: 90	*Training* 65 (N) *Test* 25 (N)	NM	Non-standard: Sound Caries	Visual Annotation: Region-based annotation under the supervision of a dentist	Smooth	NM	Yes	NM	Anterior

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TABLE 2 (Continued)

References	Population	Sample size	Train vs. test (ratio/number)	Frequency of caries	Reference standard (ICDAS vs. non-standard)	Ground truth for caries diagnosis	Tooth surface	Restoration (yes/no)	Plaque/calculus (yes/no)	Type of dentition (primary/permanent)	Tooth location
Sonavane et al. (2021)	India	Total: 74	*Training* 48 (N) *Validation* 12 (N) *Test* 14 (N)	Sound: 19 Caries: 55	Non-standard: Sound Caries	NM	NM	NM	NM	NM	NM
Kim et al. (2021)	South Korea	Images (teeth): 5251 Images (maxilla and mandible): 610 Participants: 305	*Training* 63.9% (3357) *Validation* 18.0% (946) *Test* 18.1% (948)	Sound: 4804 Occlusal caries: 415 Proximal caries: 28 Cavitation: 4	Non-Standard: Sound Non-cavitated Cavitation	Visual Annotation: One doctor of dental surgery (DDS) and one doctor of dental medicine (DMD) with more than 5 years of experience	Occlusal	NM	Yes	NM	Posterior/Anterior
Askar et al. (2021)	Germany	Images: 434 Teeth: 2781 Participants: 51	*Training* 90% *Test* 10%	Sound: 1739 (63%) Caries: 347 (13%) Fluorotic lesion: 668 (24%)	Non-standard: Sound White spot	Visual Annotation: Pixel-wise by two experts in cariology and developmental disorders	Smooth	No	Yes	NM	Anterior
Ding et al. (2021)	China	Original Images: 640 Comprehensive: 7980 Augmented: 3990 Enhance: 3990 Participants: 570	*Training* 570 (N) *Test* 70 (N)	NM	Non-standard: Primary caries presence (yes/no) Secondary caries presence (yes/No)	Visual Annotation	Occlusal, Smooth	Yes	Yes	Permanent	Posterior/Anterior
Jiang et al. (2021)	China	Images: 8544	*Training* 5987 (70%) *Validation* 1711 (20%) *Test* 865 (10%)	Mild caries: 5570 (38.1%) Moderate caries: 4370 (29.9%) Severe caries: 4670 (32.0%)	Non-Standard: Mild caries Moderate caries Severe caries	Visual/Tactile Examination: Clinical manifestations clinical diagnoses of patients Practical experience of the dentists	Occlusal	NM	Yes	NM	Posterior/Anterior

(Continues)

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TABLE 2 (Continued)

References	Population	Sample size	Train vs. test (ratio/number)	Frequency of caries	Reference standard (ICDAS vs. non-standard)	Ground truth for caries diagnosis	Tooth surface	Restoration (yes/no)	Plaque/ calculus (yes/ no)	Type of dentition (primary/ permanent)	Tooth location
Saini et al. (2021)	India	Images: 500	*Training* 400 (N) *Validation* 70 (N) *Test* 30 (N)	NM	Non-standard: Sound Caries	NM	NM	NM	NM	NM	Posterior
Zhang et al. (2022)	China	Images: 3932 Participants: 625	*Training* Images: 2507 (63.76%) Participants: 389 (62.24%) *Test* Images: 1125 (28.61%) Participants: 187 (29.92%)	Training Caries: 933 (33.24%) Test Caries: 396 (35.20%)	Merged ICDAS: Sound Carious	Visual/Tactile Examination: One dentist at a time out of three board-certified dentists	Occlusal, Smooth	NM	Yes	Permanent	Posterior/ Anterior
Kühnisch et al. (2022)	Germany	Images: 2417 Occlusal: 1317 Smooth: 1100	*Training* Total: 1891(N) Sound: 673 (N) Non-cavitated: 870 (N) Cavitated: 348 *Test* Total: 479 (N) Sound: 180 (N) Non-cavitated: 216 (N) Cavitated: 83 (N)	Sound: 853 Non-cavitated: 1086 Cavitated: 431 Excluded: 47	Merged ICDAS: 0: Sound Non-cavitated Cavitated	Visual Annotation: One experienced dentist	Occlusal, Smooth	No	No	Permanent	Posterior/ Anterior
Rashid et al. (2022)	Pakistan	Images: 90	*Training* *70%* *Test* *30%*	NM	Non-standard: Sound Caries	Visual Annotation: Two dentists with more than 3 years of experience	Smooth	NM	NM	NM	Anterior
Park et al. (2022)	South Korea	Participants:445 Images: 2348	*Training* 1636 (N) *Validation* 410 (N) *Test* 300 (N)	Carious: 1999 images	Merged ICDAS: Sound Carious	Visual Annotation: One board-certified dentist	Occlusal, Smooth	Yes	No	Permanent	Posterior/ Anterior

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TABLE 2 (Continued)

References	Population	Sample size	Train vs. test (ratio/number)	Frequency of caries	Reference standard (ICDAS vs. non-standard)	Ground truth for caries diagnosis	Tooth surface	Restoration (yes/no)	Plaque/ calculus (yes/ no)	Type of dentition (primary/ permanent)	Tooth location
Thanh et al. (2022)	Vietnam	Participants: 945 Images: 2652	*Training* 1902 *Test* 750	NM	Merged ICDAS: Sound Initial Moderate Extensive	Visual Annotation: One dentist with more than 20years of experience	Smooth	NM	Yes	Primary and permanent	Posterior/ Anterior
Khan et al. (2022)	Various population	Images: 609	*Training* *80%* *Validation* *20%*	NM	Non-standard: Sound Caries	NM	NM	NM	NM	NM	Posterior/ Anterior

Abbreviations: ICDAS, international caries detection and assessment system; N/A, not applicable; NM, not mentioned.

Regarding the establishment of the reference standard, two studies used tactile/visual examination (Jiang et al., 2021; Zhang et al., 2022), 13 studies used visual annotation from which only four reported having two or more experts (Askar et al., 2021; Kim et al., 2021; Navarro et al., 2019; Rashid et al., 2022); four studies did not report on the annotation process. Out of the 14 studies reporting on caries surfaces, three reported on occlusal surfaces, six on smooth surfaces, and five on both occlusal and smooth surfaces. Eight studies reported on the type of dentition (i.e., primary or permanent), and only two of these included primary teeth (Thanh et al., 2022; Zhang et al., 2020). Three studies trained the models exclusively for anterior teeth (Askar et al., 2021; Javid et al., 2020; Rashid et al., 2022) and three exclusively for posterior teeth (Moutselos et al., 2019; Saini et al., 2021; Yu et al., 2020; Table 2).

3.4 | Performance of AI models

Except for two studies (Moutselos et al., 2019; Zhang et al., 2020), a variety of image processing techniques were employed to enhance or augment images before applying AI models. Only four studies reported on the performance of the models for different lesion severity (e.g., cavitated or non-cavitated lesions; Kühnisch et al., 2022; Moutselos et al., 2019; Thanh et al., 2022; Wang et al., 2020); the remaining 15 studies only reported on binary detection (i.e., presence of caries lesion). Only one study reported on the performance of AI models in detecting secondary caries besides primary caries (Ding et al., 2021). There was only one study that used the non-DL approach for detecting caries lesions (Navarro et al., 2019; Table 3). Also, out of 10 studies reporting on detection tasks, six reported on IoU thresholds, which ranged from 0.4 to 0.7.

Out of the 19 articles analyzed, only 15 were used to evaluate model performances as four studies only reported accuracy. Focusing on detecting caries lesion of any severity in comparison with sound teeth, and for the classification task, the F1-scores ranged from 68.3% to 94.3%. In terms of the detection task, the F1-scores ranged from 42.8% to 95.4%. Across tasks and for detection of caries lesion in comparison to sound teeth, the F1-score for caries detection on images taken with professional cameras ranged from 68.3% to 95.4%, for images taken with intraoral cameras from 78.8% to 87.6%, and for images taken on smartphones from 42.8% to 80% (Table 3). The detailed results are laid out below.

3.4.1 | Caries detection stratified for task and image acquisition device

For classification tasks, two studies used professional cameras, which reported F1-scores of 68.3% and 89.2% (Askar et al., 2021; Kühnisch et al., 2022). Four studies used intraoral cameras of which three reported F1-scores ranging from 78.8% to 94.3% (Kim et al., 2021; Liu et al., 2020; Park et al., 2022), and one reported recall and specificity of 96.2% and 97.9% (Wang et al., 2020). One

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TABLE 3 Performance metrics of the included studies in chronological order.

References	Image acquisition device	Image preprocessing	Type of image (complete/single tooth)	AI model (DL & non-DL)	Formulation task (classification/detection/segmentation)	Sensitivity/recall (%)
Moutselos et al. (2019)	Professional camera	None	Single tooth	Mask R-CNN	Segmentation	Multi–class (ICDAS) 0: 70.6 1: 81.0 2: 27.0 3: 93.1 4: 40.0 5: 66.3 6: 82.0
Navarro et al. (2019)	Smartphone	Histogram-based contrast	Complete	SVM Decision Tree	Segmentation	NM
Zhang et al. (2020)	Smartphone	None	Complete	SSD-MobileNetV2 Faster R-CNN (ResNet50)	Detection	Binary (caries: yes) MobileNetV2: 56.2 Faster R-CNN: 71.6
Wang et al. (2020)	Intraoral Camera	NM	Single tooth	Customized CNNs	Classification	Multi–class (non-standard) sound: 96.0 white–spot: 95.0 smashed: 95.0
Liu et al. (2020)	Intraoral Camera	Retinex algorithm (Image contrast enhancement)	Complete	Mask R-CNN	Classification	Binary (caries: yes) 97.7
Yu et al. (2020)	Professional Camera	Augmentation Up–down Left–right	Complete	Control YOLOv1 YOLOv2 SSD DSSD Case custom UCDA	Detection	Binary (caries: yes) YOLOv1: 85.1 YOLOv2: 85.6 SSD: 86.7 DSSD: 87.3 UCDA: 89.8
Javid et al. (2020)	Professional camera	Sharpening filters	Complete	Mask R-CNN	Detection	Binary (caries: yes) 96.0
Sonavane et al. (2021)	NM	Horizontal flip Zooming	Single tooth	CNN (NM)	Classification	NM
Kim et al. (2021)	Intraoral Camera	Retinex algorithm (image contrast enhancement)	Complete	Classification ResNet Shufflenet V2 DenseNet ResNext Detection YOLOv3: Faster R-CNN RetinaNet	Classification Detection	Binary (caries: yes) Classification ResNext: 85.0 DenseNet: 76.6 ShuffleNet: 60.7 ResNet: 64.4

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Precision/PPV (%)	F1 (%)	Specificity (%)	Accuracy (%)	NPV (%)	AUROC (%)	IoU threshold	mAP (%)
Multi–class (ICDAS) 0: 51.4 1: 53.1 2: 62.5 3: 44.3 4: 59.3 5: 78.6 6: 92.9	Multi–class (ICDAS) 0: 59.5 1: 64.2 2: 37.7 3: 60.0 4: 47.8 5: 71.9 6: 87.1	NM	NM	NM	NM	N/A	NM
NM	NM	NM	Binary (caries: yes) SVM: 78.0 Decision Tree: 84.0	NM	NM	N/A	NM
NM	NM	NM	NM	NM	NM	0.5 and 0.75	MobileNetV2: 30.3 Faster R-CNN: 47.3
NM	NM	Multi–class (non–standard) sound: 97.9 white–spot: 95.0 smashed: 100.0	Multi–class (non-standard) sound: 95.3 white–spot: 95.3 smashed: 95.3	NM	NM	N/A	NM
Binary (caries: yes) 91.3	Binary (caries: yes) 94.3	Binary (caries: yes) 92.9	Binary (caries: yes) 95.0	NM	NM	N/A	NM
NM	NM	Binary (caries: yes) YOLOv1: 91.9 YOLOv2: 92.2 SSD: 93.4 DSSD: 94.3 UCDA: 96.1	Binary (caries: yes) YOLOv1: 89.2 YOLOv2: 90.3 SSD: 91.5 DSSD: 93.1 UCDA: 95.2	NM	NM	0.7	NM
Binary (caries: yes) 95.0	Binary (caries: yes) 95.4	NM	NM	NM	NM	0.6	NM
NM	NM	NM	Binary (caries: yes) 71.4	NM	NM	N/A	NM
Binary (caries: yes) Classification ResNext: 83.4 DenseNet: 64.0 ShuffleNet: 74.7 ResNet: 71.8	Binary (caries: yes) Classification ResNext: 84.2 DenseNet: 69.7 ShuffleNet: 67.0 ResNet: 67.9	NM	Binary (caries: yes) Classification ResNext: 96.4 DenseNet: 92.5 ShuffleNet: 93.2 ResNet: 93.1	NM	NM	NM	YOLOv3: 94.6 Faster R-CNN: 96.1 RetinaNet: 97.2

(Continues)

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TABLE 3 (Continued)

References	Image acquisition device	Image preprocessing	Type of image (complete/single tooth)	AI model (DL & non-DL)	Formulation task (classification/detection/segmentation)	Sensitivity/recall (%)
Askar et al. (2021)	Professional camera	Tooth was segmented before implementing the models Resize, input range selection, normalization	Complete	Squeeze Net	Classification	Binary (caries: yes) Tooth image: 58.3 Complete image: 58.2
Ding et al. (2021)	Smartphone	Augmentation Rotation and flipping Retinex Algorithm (Image contrast enhancement)	Complete	YOLOv3	Detection	Binary (caries: yes) Primary Caries Augmented: 49.5 Enhanced: 52.0 Comprehensive: 69.4 Secondary Caries Augmented: 52.3 Enhanced: 33.3 Comprehensive: 52.3
Jiang et al. (2021)	Smartphone Professional Camera	Enhancement (contrast and brightness) Random flipping Adding noise	Complete	Control YOLOv5s YOLOv5m YOLOv5l YOLOv3-tiny Faster R-CNN Mask R-CNN Case RDFNet	Detection	Binary (caries: yes) YOLOv5s: 57.8 YOLOv5m: 58.5 YOLOv5l: 58.2 YOLOv3-tiny: 56.6 Faster R-CNN: 82.5 Mask R-CNN: 82.4 RDFNet: 57.9
Saini et al. (2021)	NM	NM	Single tooth	VGG NET-16 VGG NET-19 INCEPTION-V3 RESNET-50	Classification	NM
Zhang et al. (2022)	Smartphone Professional Camera	Augmentation Shifting Cropping Scaling Rotation Hue Saturation Exposure	Complete	SSD-ConvNet	Classification Detection	Binary (caries: yes) Image-wise: 81.9 Box wise: 64.6

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Precision/PPV (%)	F1 (%)	Specificity (%)	Accuracy (%)	NPV (%)	AUROC (%)	IoU threshold	mAP (%)
Binary (caries: yes) Tooth image: 48.1 Complete image: 46.2	Binary (caries: yes) Tooth image: 68.3 Complete image: 68.2	Binary (caries: yes) Tooth image: 85.1 Complete image: 86.1	Binary (caries: yes) Tooth image: 84.1 Complete image: 84.1	Binary (caries: yes) Tooth image: 93.0 Complete image: 92.1	Binary (caries: yes) Tooth image: 83.1 Complete image: 83.1	N/A	NM
Binary (caries: yes) Primary Caries Augmented: 76.9 Enhanced: 81.8 Comprehensive: 93.3 Secondary Caries Augmented: 91.6 Enhanced: 100.0 Comprehensive: 100.0	Binary (caries: yes) Primary Caries Augmented: 60.0 Enhanced: 64.0 Comprehensive: 80.0 Secondary Caries Augmented: 67.0 Enhanced: 50.0 Comprehensive: 69.0	NM	NM	NM	NM	NM	Primary Caries Augmented: 55.6 Enhanced: 68.2 Comprehensive: 85.0 Secondary Caries Augmented: 56.7 Enhanced: 65.1 Comprehensive: 85.8
Binary (caries: yes) YOLOv5s: 59.7 YOLOv5m: 60.8 YOLOv5l: 60.6 YOLOv3-tiny: 58.0 Faster R-CNN: 42.1 Mask R-CNN: 42.0 RDFNet: 62.23	Binary (caries: yes) YOLOv5s: 54.3 YOLOv5m: 59.6 YOLOv5l: 59.3 YOLOv3-tiny: 57.2 Faster R-CNN: 55.7 Mask R-CNN: 55.6 RDFNet: 59.9	NM	NM	NM	NM	0.5	YOLOv5s: 55.2 YOLOv5m: 55.6 YOLOv5l: 56.3 YOLOv3-tiny: 50.0 Faster R-CNN: 42.1 Mask R-CNN: 42.0 RDFNet: 56.9
NM	NM	NM	Binary (caries: yes) VGG NET-16: 99.1 VGG NET-19: 98: 4 INCEPTIONV3: 98.9 RESNET-50: 97.4	NM	NM	N/A	NM
NM	NM	NM	NM	NM	Binary (caries: yes) Image-wise: 85.6	NM	NM

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TABLE 3 (Continued)

References	Image acquisition device	Image preprocessing	Type of image (complete/single tooth)	AI model (DL & non-DL)	Formulation task (classification/detection/segmentation)	Sensitivity/recall (%)
Kühnisch et al. (2022)	Professional Camera	Augmentation (random center and margin cropping, resizing)	Single tooth	MobileNetV2	Classification	Binary (caries: yes) 89.6 Binary (cavity: yes) 95.7
Rashid et al. (2022)	Professional Camera	Enhancement (contrast) Sharpening filters	Complete	Mask R-CNN (ResNet50)	Detection	Binary (caries: yes) 88.0
Park et al. (2022)	Intraoral Camera	Tooth was segmented using CNN (U-Net) before implementing the models Augmentation such as data shifting, image symmetry, and blurring	Complete	Classification Faster R-CNN Detection ResNet-18	Classification Detection	Binary (caries: yes) Classification 74.0 Binary (caries: yes) Detection 87.4
Thanh et al. (2022)	Smartphone	Augmentation Flipping and rotation position Enhancement Rescaling, Gaussian noise filter	Complete	YOLOv3 Faster R-CNN RetinaNet SSD	Detection	Binary (caries: yes) YOLOv3:36.9 Faster R-CNN:23.4 RetinaNet:26.5 SSD:0 Binary (cavity: yes) YOLOv3:74.0 Faster R-CNN: 71.2 RetinaNet: 63.2 SSD:26.0
Khan et al. (2022)	Various sources	Enhancement Resize, grayscale conversion Augmentation random rotation, zooming, brightness, flipping, shifting	Complete	DentalNet VGG-16 MobileNetV2 InceptionV3 ResNet18	Classification	NM

Abbreviations: AI, artificial intelligence; AUROC, area under receiver operating characteristics; CNN, convolutional neural network; ICDAS, international caries detection and assessment system; IoU, intersection over union; mAP, mean average precision; N/A, not applicable; NM, not mentioned; NPV, negative predictive value; SVM, support vector machine; UCDA, unified caries detection and assessment.

study used both smartphone and professional cameras and reach an AUROC of 85.6% (Zhang et al., 2022).

For the detection tasks, two studies used professional cameras and reported F1-scores of 86.6% and 95.4% (Javid et al., 2020; Rashid et al., 2022), and one study reported sensitivity and specificity of 89.8% and 96.1% (Yu et al., 2020). Two studies used intraoral cameras of which one reached an F1-score of 87.6% (Park et al., 2022), and the other reported mAP of 97.2%. Also, three studies used smartphones of which two reached F1-scores of 42.8% and 80.0% (Ding et al., 2021; Thanh et al., 2022), and one reported an average recall of 71.6% and mAP of 47.3%. One study used both smartphone and professional cameras and reported an F1-score of 59.9% (Jiang et al., 2021).

3.4.2 | Lesion severity stratified for task and image acquisition device

For the classification task, one study used intraoral camera and reported sensitivity and specificity of 95% and 95% for detecting white spots, respectively, and 95% and 100% for more severe caries

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Precision/PPV (%)	F1 (%)	Specificity (%)	Accuracy (%)	NPV (%)	AUROC (%)	IoU threshold	mAP (%)
Binary (caries: yes) 93.6 Binary (cavity: yes) 79.5	Binary (caries: yes) 89.2 Binary (cavity: yes) 86.8	Binary (caries: yes) 94.3 Binary (cavity: yes) 81.5	Binary (caries: yes) 92.5 Binary (cavity: yes) 93.3	Binary (caries: yes) 90.6 Binary (cavity: yes) 96.2	Binary (caries: yes) 96.4 Binary (cavity: yes) 95.5	N/A	NM
Binary (caries: yes) 85.4	Binary (caries: yes) 86.6	NM	NM	NM	NM	0.6	95.7
Binary (caries: yes) Classification 86.7 Binary (caries: yes) Detection 89.0	Binary (caries: yes) Classification 78.8 Binary (caries: yes) Detection 87.6	Binary (caries: yes) Classification 88.7	Binary (caries: yes) Classification 81.3	Binary (caries: yes) Classification 77.3	Binary (caries: yes) Classification 83.7	0.4	88.9
Binary (caries: yes) YOLOv3:49.7 Faster R-CNN:61.5 RetinaNet:61.3 SSD:0 Binary (cavity: yes) YOLOv3: 65.3 Faster R-CNN: 77.3 RetinaNet: 67.7 SSD:97.1	Binary (caries: yes) YOLOv3: 42.8 Faster R-CNN: 34.5 RetinaNet: 38.6 SSD: 0 Binary (cavity: yes) YOLOv3: 69.8 Faster R-CNN: 74.2 RetinaNet: 65.5 SSD: 43.2	Binary (caries: yes) YOLOv3:71.4 Faster R-CNN:87.7 RetinaNet:83.3 SSD:99.7 Binary (cavity: yes) YOLOv3: 86.6 Faster R-CNN:92.9 RetinaNet: 89.8 SSD:99.7	Binary (caries: yes) YOLOv3:60.7 Faster R-CNN: 67.8 RetinaNet: 65.7 SSD: 68.8 Binary (cavity: yes) YOLOv3:83.4 Faster R-CNN: 87.4 RetinaNet: 83 SSD: 81	NM	NM	NM	NM
NM	NM	NM	Binary (caries: yes) DentalNet: 90.9 VGG-16: 87.7 MobileNetV2: 84.5 InceptionV3: 82.6 ResNet18: 86.7	NM	NM	N/A	NM

lesions (Wang et al., 2020). Another study used a professional camera and reported an F1-score of 86.8% for detecting cavitated lesions compared to 89.2% for detecting any lesion (Kühnisch et al., 2022).

For the detection task, one study used smartphones and reported F-1 scores of 42.8% for white spots and 74.2% for cavitated lesions (Thanh et al., 2022). For the segmentation task, one study used professional camera and reported F1-scores of 59.5%, 64.2%, 37.7%, 60.0%, 47.8%, 71.9%, and 87.1% for ICDAS class 0–6, respectively (Moutselos et al., 2019).

3.4.3 | Controlled trials for AI models

For classification tasks, ResNext was found to have higher performance than ResNet-101, DenseNet, and SuffleNeTV2 based on the F1-score (Kim et al., 2021).

For detection tasks, in terms of localization and according to mAP, the performance of YOLOv2 was lower than SSD, and DSSD (Yu et al., 2020). The Faster R-CNN was superior to SSD-MobileNetV2 (Zhang et al., 2020). The performance of Faster

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R-CNN and RetinaNet was higher than YOLOv3 (Kim et al., 2021). However, in another study, the performance of Faster R-CNN and Mask R-CNN was lower compared to the YOLOv3-tiny and YOLOv5 and RDFNet (Jiang et al., 2021). Based on the results, it seems the two-stage models (i.e., R-CNN) do not have an advantage over one-stage models. For the class probability of the boxes and according to F1- score, the results were again equivocal with one study finding the performance of Faster R-CNN and Mask R-CNN lower than YOLOv3-tiny,YOLOv5 and RDFNet (Jiang et al., 2021), while another study reporting that the performance of Faster R-CNN was higher than YOLOv3, RetinaNet, and SSD (Thanh et al., 2022).

3.4.4 | Subgroup analysis for caries type

The only study that compared the performance of AI models between primary and secondary caries found that the models were comparable in terms of mAP (primary: 85.0% vs. secondary: 85.8%), but significantly different in terms of F1-score (primary: 80.0% vs. secondary: 69.0%) (Ding et al., 2021).

4 | DISCUSSION

The present systematic review evaluated the performance of AI-based models for detecting dental caries on oral photographs. To our knowledge, this is the first review to evaluate the application of DL and non-DL models for detecting the presence and severity of caries considering different photograph modalities, analysis tasks, and AI algorithms. Overall, although the performance metrics showed variability among studies due to methodological heterogeneity, AI-based models seem to be promising for detecting dental caries on oral photographs.

The analysis revealed that among the included studies, only a limited number featured a control design that either exclusively controlled for the AI models (Jiang et al., 2021; Kim et al., 2021; Saini et al., 2021; Thanh et al., 2022; Zhang et al., 2020) or caries type (i.e., primary vs. secondary) (Ding et al., 2021). Consequently, only these studies were able to adjust for potential confounders in the assessment of AI models. The main finding was that two-stage models (e.g., R-CNN) did not exhibit a substantial advantage over one-stage models (e.g., YOLOv). This observation could be attributed to several factors, including overfitting resulting from limited sample sizes, redundant complexity in the two-stage models, and data quality. However, additional trials are required to substantiate these findings. Regarding caries type, the AI models demonstrated superior performance in detecting primary caries compared to secondary caries. This difference is likely due to the variability in the clinical presentation of secondary caries (e.g., amalgam, composites) and the smaller number of labeled secondary caries in the dataset (Ding et al., 2021).

Except for the aforementioned comparative studies, the methodological heterogeneity precluded the consolidation and direct comparison of results from the included studies. However, stratification of the findings according to photographic modalities and formulation tasks suggested that AI models have superior performance when utilizing professional and intraoral cameras as opposed to smartphone images. The detection of dental caries in oral photographs captured by professional and intraoral cameras can significantly aid oral health practitioners in mitigating diagnostic discrepancies by objectively verifying the presence or severity of caries (Kühnisch et al., 2022; Schwendicke et al., 2020). Although smartphone cameras may not offer the same image quality as their professional or intraoral counterparts, their widespread availability, cost-effectiveness, and applicability in teledentistry and dental public health surveys make them promising sensors. Our review has highlighted the need for further work in this area and for further evaluation of AI models using smartphone-derived imagery.

The early-stage application of AI in dental caries detection is evident from the publication dates of included studies and the methodological limitations, which must be considered when interpreting results. Firstly, only two studies utilized the current clinical best practice of visual-tactile examination for establishing the reference test (Jiang et al., 2021; Zhang et al., 2022) while other studies relied on visual annotations. Secondly, the limited data available in dentistry restricts AI model performance; while some studies employ augmentation techniques to increase the sample size (Kühnisch et al., 2022; Zhang et al., 2022), training models on limited data remains an active research area in deep learning. Thirdly, the single-center nature of most studies challenges model robustness and generalizability, with no comprehensive demographic or socioeconomic data provided. Fourth, only two studies evaluated segmentation models (Moutselos et al., 2019; Navarro et al., 2019) while the others focused on classification or detection, likely due to the cost and difficulty of pixel-wise expert annotations.

The main limitation of the methods used thus far is their inability to detect disease activity, which is an important criterion for treatment decisions. Using the current tools may lead to overdiagnosis and overtreatment, which may lead to unnecessary expenses, health loss, reduced accessibility of care and overall increased oral healthcare inequalities. Caution is advised when advocating for the use of AI-based models in their current form, as the field is still in its early stages. That said, automatic caries screening could support a transition of long-interval visits or visits only based on disease symptoms to real-time monitoring. The resultant increased increase self-awareness of oral health leading to timely visits to dentists and may reduce disease burden and severity. Concurrently, real-time surveillance would likely increase the false positive rate and consequently individual anxiety as well as healthcare system utilization. Further study is necessary to estimate the benefits of earlier detection against the harms of such false positive diagnoses.

5 | CONCLUSIONS

The number of AI models for screening dental caries from oral photographs has increased steadily over the last few years. Despite the

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heterogeneity in methodology and variability of performance metrics, the application of AI models seems promising for caries detection from oral photographs. Future research in AI-based dental caries detection should address the methodological limitations observed in the current review. Controlled trials should be conducted more broadly, accounting for potential confounding factors. Increased transparency regarding recording device parameters and processing approaches would improve understanding of study results. Exploring the potential of smartphone cameras for dental caries detection in various settings should be considered. Lastly, future research must consider confounding factors, such as the settings and photographer (e.g., patient, clinician, researcher), along with other factors including preprocessing techniques, sample size, ground truth, region of interest, and caries frequency, which may impact the results.

AUTHOR CONTRIBUTIONS

Mohammad Moharrami: Conceptualization; methodology; data curation; writing – original draft; visualization; software. **Julie Farmer:** Data curation; formal analysis; methodology; software. **Sonica Singhal:** Writing – review and editing; supervision; investigation. **Erin Watson:** Writing – review and editing; supervision; investigation. **Michael Glogauer:** Writing – review and editing; resources; formal analysis; investigation. **Alistair Johnson:** Writing – review and editing; methodology. **Falk Schwendicke:** Writing – review and editing; visualization; validation; formal analysis; conceptualization. **Carlos Quinonez:** Conceptualization; resources; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

All authors have no conflicts of interest to disclose.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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APPENDIX 1

TABLE A1 Modified signalling questions of QUADAS-2 for critical appraisal.

Domain	Leading question
Patient selection	Were data imbalances (if there were any) addressed in the study?
	Did the study avoid inappropriate data exclusion? (e.g., difficult to diagnose or considered as outliers without a specific outlier detection method)
	Was a consecutive or random sample of patients (or data) enrolled?
	Was the test dataset separate from the training and validation datasets?
	Applicability: Are there concerns that the included data and setting did not match the review question?
Index test	Were the deep learning and machine learning results interpreted without knowledge of the results of the reference standard?
	If a threshold was used, was it pre-specified?
	Was the method described in sufficient detail to reproduce the presented results?
	Did the study perform any robustness or sensitivity analysis of their model?
	Applicability: Are there concerns that the method, its conduct, or interpretation differed from the review question?
Reference standard	Were the reference standard results interpreted without knowledge of the results of the index test?
	Did the study use a gold standard defined?
	If not, was the annotation procedure described in the study and found to minimize bias?
	Did the study sufficiently report its limitations, biases, or issues around generalizability?
	Applicability: Are there concerns that the target condition as defined by the reference standard did not match the question?
Flow and timing	Were all data included in the analysis?
	Was there an appropriate interval between the index test and reference standard?
	Did all data have a reference standard?
	Did all data have the exact reference standard?

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